

Camera Ready Summary

Conference Name

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Paper ID

184

Paper Title

A Deep Learning Model for Image Super-Resolution and Colorization

Abstract

This study introduces a deep learning-based framework for automatic image colorization, which aims to generate realistic color representations from grayscale input images. Leveraging the power of Convolutional Neural Networks (CNNs), our model is designed to learn the complex mapping between luminance and chrominance components in natural images. The architecture adopts an encoder-decoder structure enriched with skip connections to effectively capture both global semantic information and fine-grained spatial details. Training is conducted on large-scale color image datasets, including ImageNet and Places, enabling the network to generalize across a wide range of scene types and object categories. Quantitative evaluation using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) indicates strong performance relative to existing methods. Qualitative results further confirm the model's ability to produce vibrant and semantically coherent colorizations. Due to its compact design and end-to-end trainability, the proposed system holds promise for practical deployment in domains such as historical image restoration, content creation, and digital entertainment.

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A_Deep_Learning_Model_for_Image_Super_Resolution_and_Colorization.pdf (1.9 Mb, 11/15/2025, 3:46:40 PM)

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Camera Ready Questions Response

1. Name of Presenter

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3. Mode of Presentation

Online

4. Originality and AI content

Agreement accepted
